

Hearth Extensions and Clearances to Combustibles

How far anything that can burn must stay from the fire. One threshold decides the whole hearth question.

The hearth extension is the protected floor in front of the opening, there to catch embers and radiant heat. How far it must reach depends on one number: the size of the firebox opening.

The numbers here

Hearth extension — minimum thickness of solid masonry or concrete	4 in
Firebox opening threshold	6 sq ft
Opening of 6 sq ft or less — extension in front	16 in
Opening of 6 sq ft or less — extension to each side	8 in
Opening larger than 6 sq ft — extension in front	20 in
Opening larger than 6 sq ft — extension to each side	12 in

YOU HAVE MISSED THIS 4 TIMES

A fireplace hearth should be made of solid masonry or concrete that is at least four inches thick and _____ inches deep.

16.

YOU HAVE MISSED THIS 2 TIMES

If the masonry fireplace firebox opening is _____, the hearth extension should extend at least sixteen inches in front and at least eight inches to each side.

Six square feet or less.

Sixteen and eight for a small opening, twenty and twelve for a large one. The bigger the fire, the further the embers travel — the numbers are not arbitrary and they step up together.

Clearances

The numbers here

Mantel clearance from the fireplace opening	6 in
Clearance to combustibles — interior masonry chimney	2 in
Metal chimney clearance to combustibles, including insulation	2 in

Clearance reduction devices

YOU HAVE MISSED THIS 4 TIMES

The manufacturer of a wood-burning stove recommends a minimum of twelve inches to combustibles. What is the minimum clearance using a sheet metal and air space clearance reduction device?

Twelve inches.

Four times missed, and it is a trap about authority rather than about heat. A reduction device can cut a clearance set by code, but it cannot override the manufacturer's own listed instruction. The manufacturer said twelve, so it stays twelve. Whenever a question sets a manufacturer's figure against a general rule, the manufacturer wins.

Say it back

Cover the page. Answer each one out loud.

1. Minimum hearth thickness, and the opening size that splits the two cases?
2. Extension front and side for a small opening, and for a large one?
3. Mantel clearance from the opening?
4. Clearance to combustibles for an interior masonry chimney and for a metal chimney?
5. Manufacturer says twelve inches; you fit a reduction device. What is the clearance now, and why?